



南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

Tutorial Questions (Week 6)

(For lecture week 3)

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Outline

- Review
- Basic Problems with Answers 2.3, 2.7, 2.13
- Basic Problems 2.24, 2.26
- Q&A

Basic knowledge review

- CT/DT LTI systems
- Convolution operation procedure
 1. **Figure computation** based on “Flip-slide-multiply-sum/integral”
 2. Some known or **typical convolution results**
 3. **Properties** of convolution
- Unit impulse response and properties of LTI systems
 1. Memoryless or with memory
 2. Causality
 3. Invertibility
 4. Stability

If we have $x_k[n] \rightarrow y_k[n]$, then

$$x[n] = \sum_k a_k x_k[n] \xrightarrow{\text{Linear System}} y[n] = \sum_k a_k y_k[n]$$

$$x[n] = \dots + x[-2]\delta[n+2] + x[-1]\delta[n+1] + x[0]\delta[n] + x[1]\delta[n-1] + \dots$$

$\Downarrow$

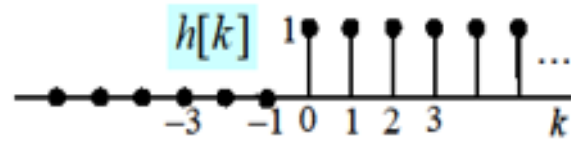
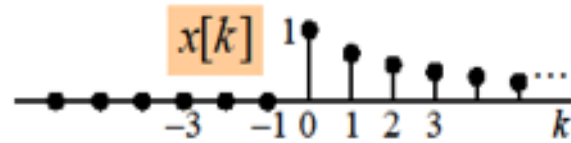
$$x[n] = \sum_{k=-\infty}^{+\infty} \underbrace{x[k]}_{\text{Coefficients}} \underbrace{\delta[n - k]}_{\text{Basic Signals}}$$

Important to note
the “-” sign

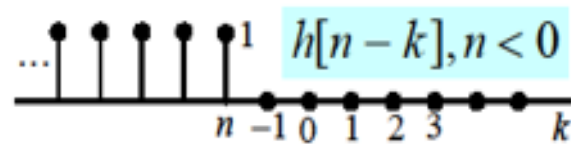
$$x[n] = \sum_{k=-\infty}^{+\infty} x[k] \delta[n - k] \longrightarrow y[n] = \underbrace{\sum_{k=-\infty}^{+\infty} x[k] h[n - k]}_{\text{convolution sum}} = x[n] * h[n]$$

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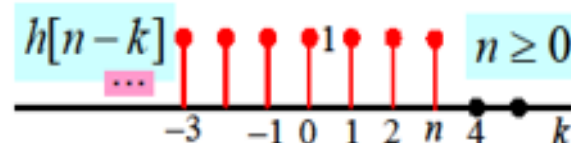
Example 1



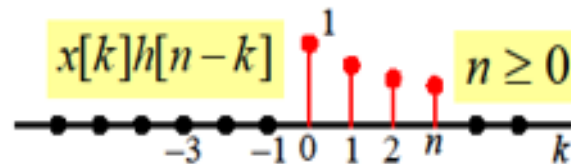
Flip



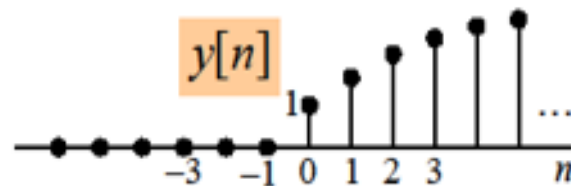
Slide



Multiply



Sum



$$\triangleright x[n] = a^n u[n]$$

$$\triangleright h[n] = u[n]$$

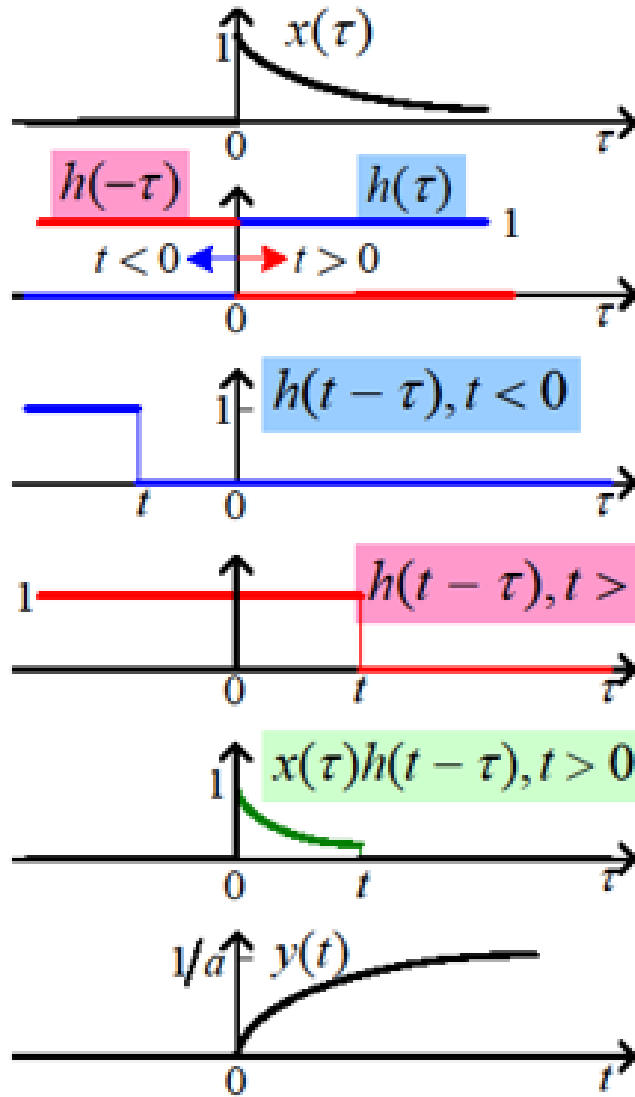
$$\triangleright y[n] = x[n] * h[n] ?$$

$$h[n-k]$$

$$x[k]h[n-k] = \begin{cases} a^k, & 0 \leq k \leq n \\ 0, & k < 0, k > n \end{cases}$$

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k] = \sum_{k=0}^n a^k = \frac{1-a^{n+1}}{1-a}$$

Example 2

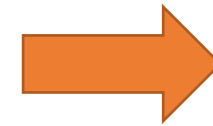


$$x(t) = e^{-at}u(t)$$

$$h(t) = u(t)$$

$$y(t) = x(t) * h(t)$$

$$y(t) = \begin{cases} \frac{1-e^{-at}}{a}, & t \geq 0 \\ 0, & t < 0 \end{cases}$$



$$\frac{1-e^{-at}}{a}u(t)$$

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$$x(t) * h(t) = \int_{-\infty}^{+\infty} x(\tau)h(t - \tau) d\tau$$

$$x[n] * h[n] = \sum_{k=-\infty}^{+\infty} x[k]h[n - k]$$

表 3.4 基本信号的卷积表

连续时间卷积积分			离散时间卷积和		
$x(t)$	$h(t)$	$x(t) * h(t)$	$x[n]$	$h[n]$	$x[n] * h[n]$
$x(t)$	$\delta(t)$	$x(t)$	$x[n]$	$\delta[n]$	$x[n]$
$x(t)$	$u(t)$	$\int_{-\infty}^t x(\tau) d\tau$	$x[n]$	$u[n]$	$\sum_{k=-\infty}^n x[k]$
$x(t)$	$\delta'(t)$	$x'(t)$	$x[n]$	$\Delta\delta[n]$	$x[n] - x[n-1]$
$u(t)$	$u(t)$	$tu(t)$	$u[n]$	$u[n]$	$(n+1)u[n]$
$e^{-at}u(t)$	$u(t)$	$\frac{1-e^{-at}}{a}u(t)$	$a^n u[n]$	$u[n]$	$\frac{1-a^{n+1}}{1-a}u[n]$
$\sin(\omega t)u(t)$	$u(t)$	$\frac{1-\cos(\omega t)}{\omega}u(t)$	$\sin(\Omega n)u[n]$	$u[n]$	
$\cos(\omega t)u(t)$	$u(t)$	$\frac{\sin(\omega t)}{\omega}u(t)$	$\cos(\Omega n)u[n]$	$u[n]$	
$e^{-at}u(t)$	$e^{-at}u(t)$	$te^{-at}u(t)$	$a^n u[n]$	$a^n u[n]$	$(n+1)a^n u[n]$
$e^{-at}u(t)$	$e^{-bt}u(t)$	$\frac{e^{-at} - e^{-bt}}{b-a}u(t)$	$a^n u[n]$	$b^n u[n]$	$\frac{b^{n+1} - a^{n+1}}{b-a}u[n]$

说明：表 3.4 中空着的卷积和运算结果，感兴趣的读者可自行补上。

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Properties of convolution

□ Commutative property

$$x(t) * h(t) = h(t) * x(t)$$

$$\int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau = \int_{-\infty}^{\infty} h(\tau)x(t-\tau)d\tau$$

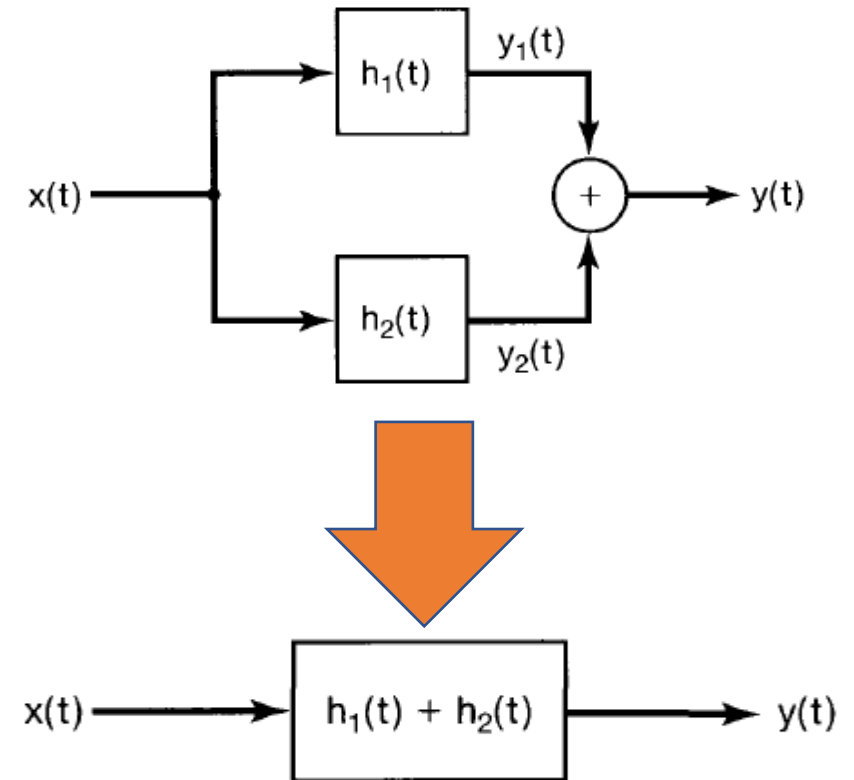
$$x[n] * h[n] = h[n] * x[n]$$

$$\sum_{m=-\infty}^{\infty} x[m]h[n-m] = \sum_{m=-\infty}^{\infty} h[m]x[n-m]$$

□ Distributive property

$$x(t) * [h_1(t) + h_2(t)] = x(t) * h_1(t) + x(t) * h_2(t)$$

$$x[n] * \{h_1[n] + h_2[n]\} = x[n] * h_1[n] + x[n] * h_2[n]$$



Properties of convolution

□ Associative property

$$[x(t) * h_1(t)] * h_2(t) = x(t) * [h_1(t) * h_2(t)]$$

$$\{x[n] * h_1[n]\} * h_2[n] = x[n] * \{h_1[n] * h_2[n]\}$$

□ Time-invariant property (Collect time shift)

$$y(t) = x(t) * h(t)$$

$$x(t) * h(t - t_0) = y(t - t_0)$$

$$x(t - t_1) * h(t - t_2) = y(t - t_1 - t_2)$$

$$x[n] * h[n] = y[n]$$

$$x[n] * h[n - m] = y[n - m]$$

$$x[n - m_1] * h[n - m_2] = y[n - m_1 - m_2]$$

Properties of convolution

□ Difference property

$$\frac{d}{dt} [x(t) * h(t)] = x(t) * \frac{dh(t)}{dt} = \frac{dx(t)}{dt} * h(t) = \frac{dy(t)}{dt}$$

$$\nabla \{x[n] * h[n]\} = \nabla x[n] * h[n] = x[n] * \nabla h[n] = \nabla y[n]$$

□ Integral property

$$\int_{-\infty}^t [x(\tau) * h(\tau)] d\tau = x(t) * \int_{-\infty}^t h(\tau) d\tau = \int_{-\infty}^t x(\tau) d\tau * h(t) = \int_{-\infty}^t y(\tau) d\tau$$

$$\sum_{k=-\infty}^n \{x[k] * h[k]\} = x[n] * \left\{ \sum_{k=-\infty}^n h[k] \right\} = \left\{ \sum_{k=-\infty}^n x[k] \right\} * h[n] = \sum_{k=-\infty}^n y[k]$$

Properties of convolution

□ For unit impulse/step signal

□ More unit impulse/step signals, more simple

$$x(t) * \delta(t) = x(t)$$

$$x(t) * \delta(t - t_0) = x(t - t_0)$$

$$x(t) * u(t) = \int_{-\infty}^t x(\tau) d\tau$$

$$u(t) = \int_{-\infty}^t \delta(\tau) d\tau$$

$$x[n] * \delta[n] = x[n]$$

$$x[n] * \delta[n - m] = x[n - m]$$

$$x[n] * u[n] = \sum_{m=-\infty}^n x[m]$$

Causality

$$h[n] = 0 \quad \text{for all } n < 0$$

Memoryless

$$h[n] = K\delta[n] \quad h(t) = K\delta(t)$$

$$y[n] = Kx[n] \quad y(t) = Kx(t)$$

Stability

$$\sum_{k=-\infty}^{+\infty} |h[k]| < \infty$$

Invertibility

$$h_1[n] * h_2[n] = \delta[n]$$

Basic Problems with Answers 2.3

2.3. Consider an input $x[n]$ and a unit impulse response $h[n]$ given by

$$x[n] = \left(\frac{1}{2}\right)^{n-2} u[n-2],$$

$$h[n] = u[n+2].$$

Determine and plot the output $y[n] = x[n] * h[n]$.

2.3 Let us define the signals

$$x_1[n] = \left(\frac{1}{2}\right)^n u[n]$$

and

$$h_1[n] = u[n].$$

We note that

$$x[n] = x_1[n-2] \quad \text{and} \quad h[n] = h_1[n+2]$$

Now,

$$y[n] = x[n] * h[n] = x_1[n-2] * h_1[n+2] = \sum_{k=-\infty}^{\infty} x_1[k-2] h_1[n-k+2]$$

By replacing k with $m+2$ in the above summation, we obtain

$$y[n] = \sum_{m=-\infty}^{\infty} x_1[m] h_1[n-m] = x_1[n] h_1[n]$$

Using the results of Example 2.1 in the text book, we may write

$$y[n] = 2 \left[1 - \left(\frac{1}{2}\right)^{n+1} \right] u[n]$$

Basic Problems with Answers 2.7

2.7. A linear system S has the relationship

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]g[n-2k]$$

between its input $x[n]$ and its output $y[n]$, where $g[n] = u[n] - u[n-4]$.

- (a) Determine $y[n]$ when $x[n] = \delta[n-1]$.
- (b) Determine $y[n]$ when $x[n] = \delta[n-2]$.
- (c) Is S LTI?
- (d) Determine $y[n]$ when $x[n] = u[n]$.

2.7.(a) Given that

$$x[n] = \delta[n-1],$$

we see that

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]g[n-2k] = g[n-2] = u[n-2] - u[n-6]$$

(b) Given that $x[n] = \delta[n-2]$,

we see that $y[n] = \sum_{k=-\infty}^{\infty} x[k]g[n-2k] = g[n-4] = u[n-4] - u[n-8]$

(c) The input to system in part(b) is the same as the input in part(a) shifted by 1 to the right. If S is the time invariant then the system output obtained in part(b) has to be the same as the system output obtained in part (a) shifted by 1 to the right. Clearly, this is not the case. Therefore, the system is not LTI.

(d) If $x[n] = u[n]$, then

$$\begin{aligned} y[n] &= \sum_{k=-\infty}^{\infty} x[k]g[n-2k] \\ &= \sum_{k=-\infty}^{\infty} g[n-2k] \end{aligned}$$

The signals $g[n-2k]$ is plotted for $k=0,1,2$ in Figure S2.7. From this figure it is clear that

$$y[n] = \begin{cases} 1, & n = 0, 1 \\ 2, & n > 1 \\ 0, & \text{otherwise} \end{cases} = 2u[n] - \delta[n] - \delta[n-1]$$

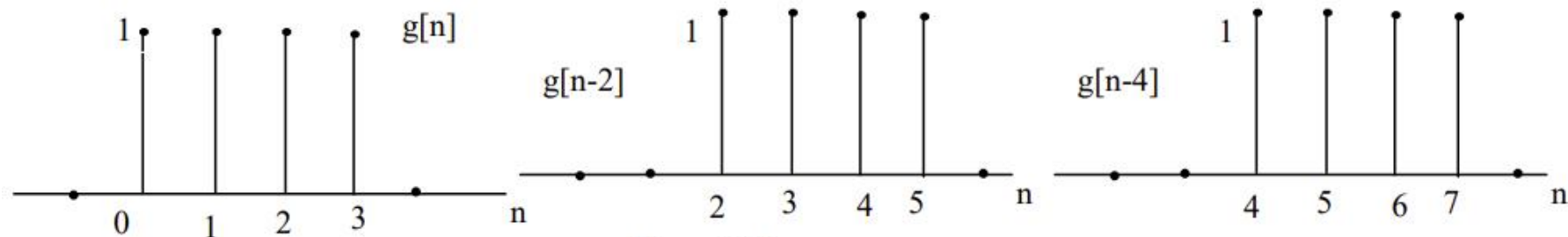


Figure S2.7

Basic Problems with Answers 2.13

2.13. Consider a discrete-time system S_1 with impulse response

$$h[n] = \left(\frac{1}{5}\right)^n u[n].$$

- (a) Find the integer A such that $h[n] - Ah[n - 1] = \delta[n]$.
- (b) Using the result from part (a), determine the impulse response $g[n]$ of an LTI system S_2 which is the inverse system of S_1 .

2.13. (a) we require that

$$\left(\frac{1}{2}\right)^n u[n] - A \left(\frac{1}{2}\right)^{n-1} u[n-1] = \delta[n]$$

Putting $n=1$ and solving for A gives $A = \frac{1}{5}$

(b) From part (a), we know that

$$h[n] - \frac{1}{5} h[n-1] = \delta[n]$$

$$h[n] * \left(\delta[n] - \frac{1}{5} \delta[n-1]\right) = \delta[n]$$

From the definition of an inverse system, we may argue that

$$g[n] = \delta[n] - \frac{1}{5} \delta[n-1]$$

Basic Problems 2.24

2.24. Consider the cascade interconnection of three causal LTI systems, illustrated in Figure P2.24(a). The impulse response $h_2[n]$ is

$$h_2[n] = u[n] - u[n - 2], \quad \longrightarrow \delta[n] + \delta[n - 1]$$

and the overall impulse response is as shown in Figure P2.24(b).

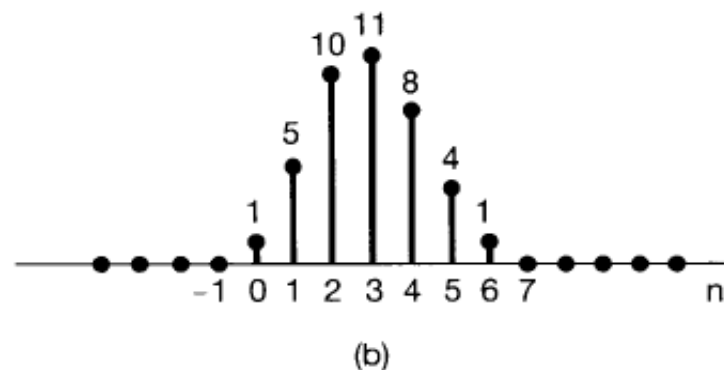
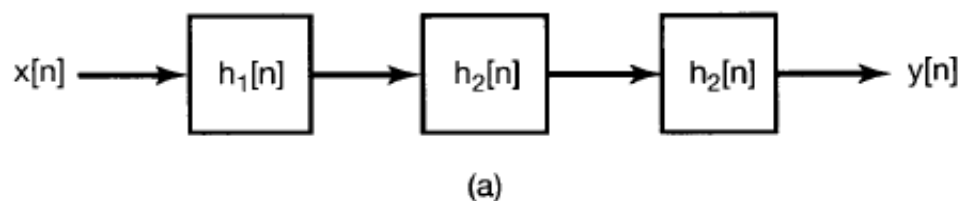


Figure P2.24

- (a) Find the impulse response $h_1[n]$.
- (b) Find the response of the overall system to the input

$$x[n] = \delta[n] - \delta[n - 1].$$

2.24 (a) we have given that $h_2 = \delta(n) + \delta(n-1)$. Therefore,

$$h_2[n] * h_2[n] = \delta[n] + 2\delta[n-1] + \delta[n-2]$$

since

$$h[n] = h_1[n] * [h_2[n] * h_2[n]]$$

we get

$$h[n] = h_1[n] + 2h_1[n-1] + h_1[n-2]$$

Therefore

$$h[0] = h_1[0] \Rightarrow h_1[0] = 1$$

$$h[1] = h_1[1] + 2h_1[0] \Rightarrow h_1[1] = 3$$

$$h[2] = h_1[2] + 2h_1[1] + h_1[0] \Rightarrow h_1[2] = 3$$

$$h[3] = h_1[3] + 2h_1[2] + h_1[1] \Rightarrow h_1[3] = 2$$

$$h[4] = h_1[4] + 2h_1[3] + h_1[2] \Rightarrow h_1[4] = 1$$

$$h[5] = h_1[5] + 2h_1[4] + h_1[3] \Rightarrow h_1[5] = 0.$$

$h_1[n] = 0$ for $n < 0$ and $n \geq 5$.

(b) in this case

$$Y[n] = x[n] * h[n] = h[n] - h[n-1].$$

Basic Problems 2.26

2.26. Consider the evaluation of

$$y[n] = x_1[n] * x_2[n] * x_3[n],$$

where $x_1[n] = (0.5)^n u[n]$, $x_2[n] = u[n + 3]$, and $x_3[n] = \delta[n] - \delta[n - 1]$.

- (a) Evaluate the convolution $x_1[n] * x_2[n]$.
- (b) Convolve the result of part (a) with $x_3[n]$ in order to evaluate $y[n]$.
- (c) Evaluate the convolution $x_2[n] * x_3[n]$.
- (d) Convolve the result of part (c) with $x_1[n]$ in order to evaluate $y[n]$.

2.26 (a) we have

$$\begin{aligned} y_1[n] &= x_1[n] * x_2[n] = \sum_{k=-\infty}^{\infty} x_1[k] x_2[n-k] \\ &= \sum_{k=0}^{\infty} (0.5)^k u[n+3-k] \end{aligned}$$

This evaluates to

$$y_1[n] = x_1[n] * x_2[n] = \begin{cases} 2[1 - (1/2)^{n+4}], & n \geq -3 \\ 0, & \text{otherwise} \end{cases}$$

(b) now

$$y[n] = x_3[n] * y_1[n] = y_1[n] - y_1[n-1]$$

Therefore

$$y[n] = \begin{cases} 2(1 - (1/2)^{n+3}) + 2(1 - (1/2)^{n+4}) = (1/2)^{n+3}, & n \geq -2 \\ 1, & n = -3 \\ 0, & \text{otherwise} \end{cases}$$

$$\text{Therefore, } y[n] = (1/2)^{n+3} u[n+3].$$

(c) we have

$$y_2[n] = x_2[n] * x_3[n] = u[n+3] - u[n+2] = \delta[n+3]$$

(d) from the result of part (c), we get

$$Y[n] = y_2[n] * x_1[n] = x_1[n+3] = (1/2)^{n+3} u[n+3]$$



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Thanks for Your Attendance

Q&A